

Session 06

Introduction to Information Systems



KHOA CÔNG NGHỆ THÔNG TIN
FACULTY OF INFORMATION TECHNOLOGY
UNIVERSITY OF SCIENCE - HO CHI MINH CITY

Learning Objective

- Knowing the potential impact of information systems (IS) in business
- Understanding various aspects of IS:
 - Describe the components of computer-based information systems.
 - Describe the various types of information systems.
 - Identify the major information systems that support each organizational level.
 - Describe the fundamental of database technology
 - Describe how information resources are managed and discuss the roles of the information systems department and the end users.
- Knowing the training program of Information Systems Major

- ☐ Why learn about Information Systems
- ☐ What is an Information System
- ☐ Business Information Systems
- ☐ Database Fundamentals
- ☐ Careers in Information Systems
- ☐ Information Systems Major

Why Learn About Information System?

Why Learn About IS?

- There are many examples of the ways in which information systems and technologies are embedded in your lives, examples:
 - Register for classes
 - Take classes, and not just classes from your university
 - Access class syllabi, information, PowerPoints, and lectures
 - Research class papers and presentations
 - Conduct banking
 - Pay your bills
 - Research, shop, and buy products from companies or other people
 - Sell your “stuff ”
 - Search for, and apply for, jobs
 - Make your travel reservations (hotel, airline, rental car)

Why Learn About IS?

- Information systems are used in almost every profession, including:
 - Entrepreneurs
 - Small, medium to large multinationals
 - Production, finance, accounting, marketing, management

- Information systems are indispensable to help you to achieve your career goals

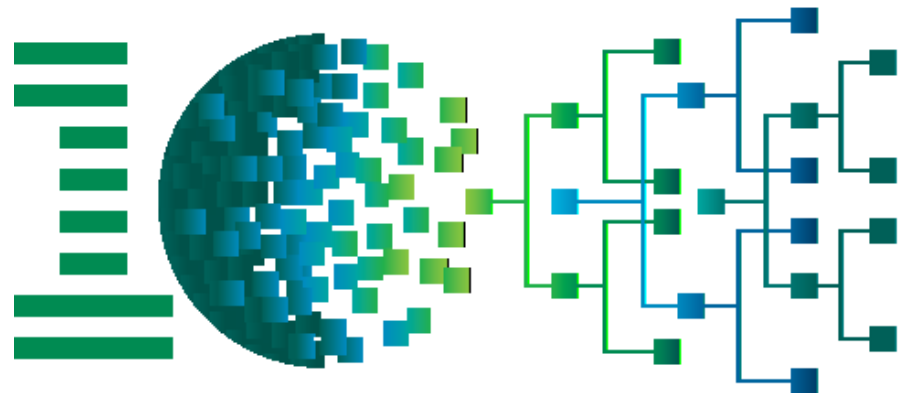
Why Learn About IS?

Position	Job Description
Chief Information Officer	Highest-ranking IS manager; is responsible for all strategic planning in the organization
IS Director	Manages all systems throughout the organization and the day-to-day operations of the entire IS organization
Information Center Manager	Manages IS services such as help desks, hot lines, training, and consulting
Applications Development Manager	Coordinates and manages new systems development projects
Project Manager	Manages a particular new systems development project
Systems Manager	Manages a particular existing system
Operations Manager	Supervises the day-to-day operations of the data and/or computer center
Programming Manager	Coordinates all applications programming efforts
Systems Analyst	Interfaces between users and programmers; determines information requirements and technical specifications for new applications
Business Analyst	Focuses on designing solutions for business problems; interfaces closely with users to demonstrate how IT can be used innovatively
Systems Programmer	Creates the computer code for developing new systems software or maintaining existing systems software
Applications Programmer	Creates the computer code for developing new applications or maintaining existing applications
Emerging Technologies Manager	Forecasts technology trends; evaluates and experiments with new technologies
Network Manager	Coordinates and manages the organization's voice and data networks
Database Administrator	Manages the organization's databases and oversees the use of database-management software
Auditing or Computer Security Manager	Oversees the ethical and legal use of information systems
Webmaster	Manages the organization's World Wide Web site
Web Designer	Creates World Wide Web sites and pages

What is an Information System?

What is an information system?

A set of interrelated *components* that
collect/retrieve, process, store and distribute
information
to *support decision making and control* in an
organization

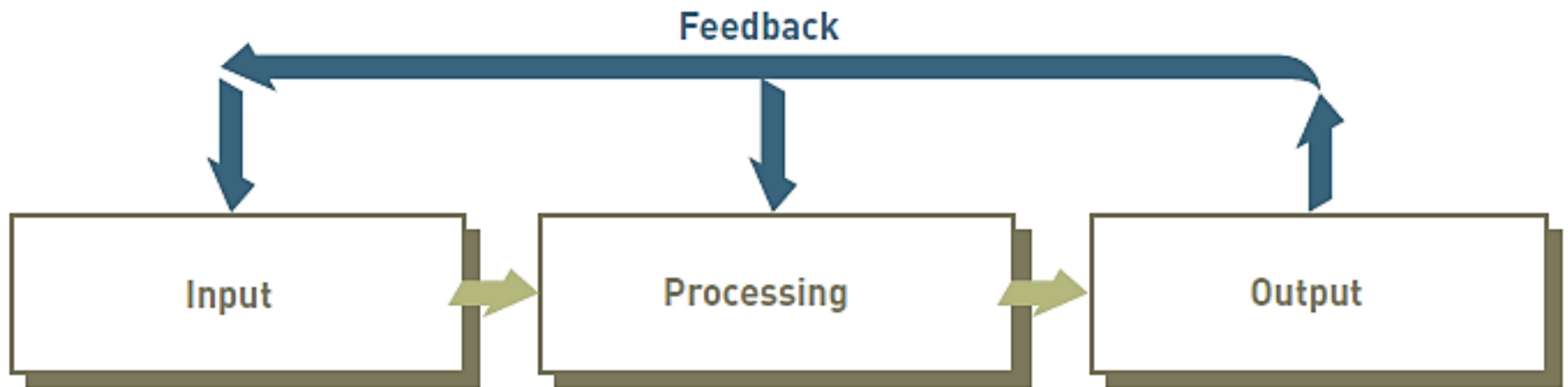


From ibmims

What is an Information System?

- An IS is a set of interrelated elements that:
 - Collect (input)
 - Manipulate (process)
 - Store
 - Disseminate (output) data and information
 - Provide a corrective reaction (feedback mechanism) to meet an objective

Components of an IS



Input, Processing, Storing, Output, Feedback

- Input: activity of gathering and capturing raw data
 - By person using a keyboard, scanner, mouse...
 - By computer, device, sensor (video, images, ...)
- Processing: converting data into useful outputs
 - i.e. purchase order □ inventory reports
- Storing: computer data storage
 - Storing data for calculation
- Output: production of useful information
 - i.e. reports, documents, charts, graphs, pictures, etc.
- Feedback: information from the system
 - Used to make changes to input or processing activities

Concepts of Information System

- ☐ Data, Information and Knowledge
- ☐ Business process
- ☐ Information system development process
- ☐ Information system development tools

Data, Information, and Knowledge

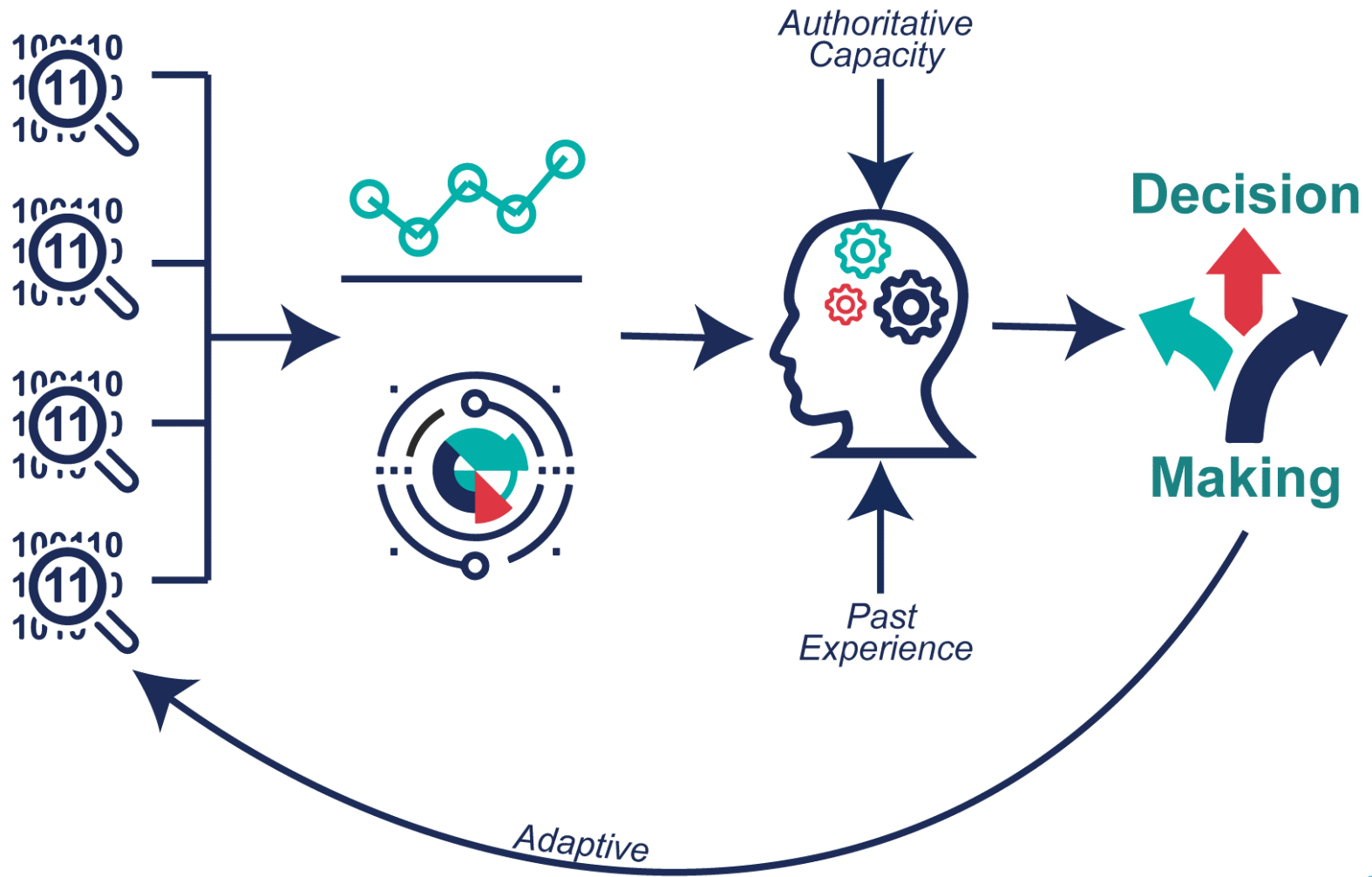
- Data: raw facts
- Information: collection of facts organized in such a way that they have value beyond the facts themselves
- Knowledge: awareness and understanding of someone or something which is acquired through experience or education by perceiving, discovering, or learning

Data, Information, Knowledge

Data
Raw

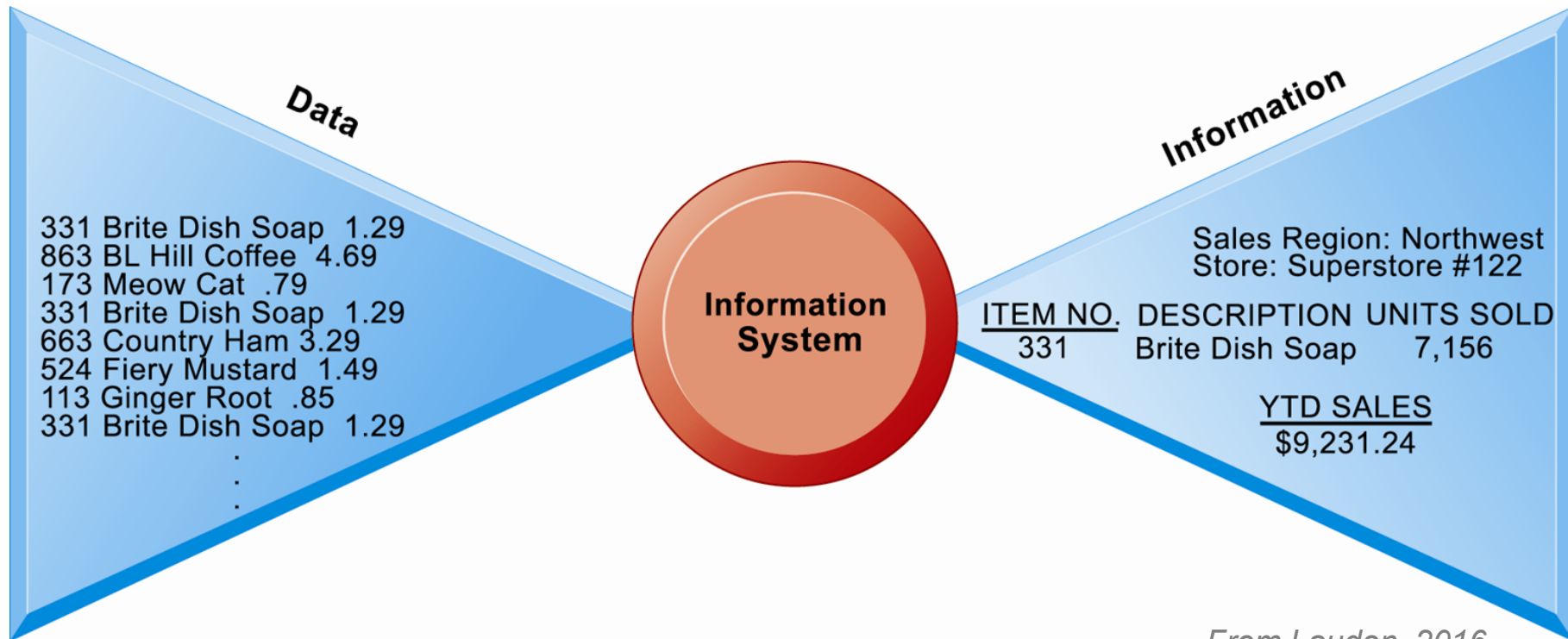
Information
Processed

Knowledge
Actionable



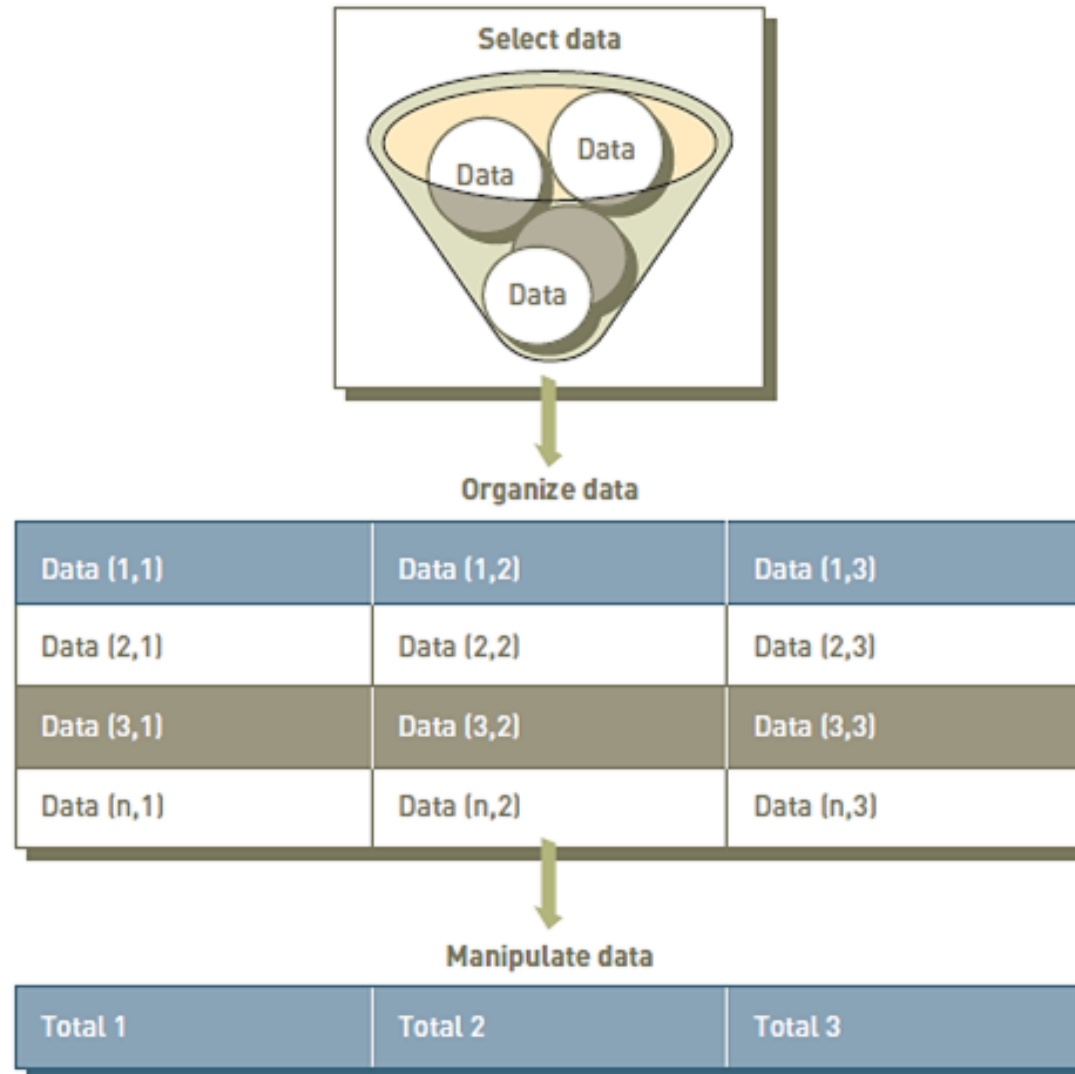
Data and Information

- Data are streams of raw facts
- Information is data shaped into meaningful form



From Laudon, 2016

Process of Transforming Data into Information



- ☐ Which of the following is NOT an example of data?
- A. 231456
 - B. 32, CMT8, P6, Tan Binh
 - C. Blue
 - D. B, C

- ☐ Which of the following is an example of information?
- A. 32, CMT8, P6, Tan Binh
 - B. Address
 - C. Address of my friend
 - D. B, C

- ☐ Which of the following is correct?
- A. Data leads to information, and information leads to knowledge.
 - B. Knowledge leads to information, and information leads to data
 - C. Data leads to knowledge and knowledge leads to information
 - D. A, B, C

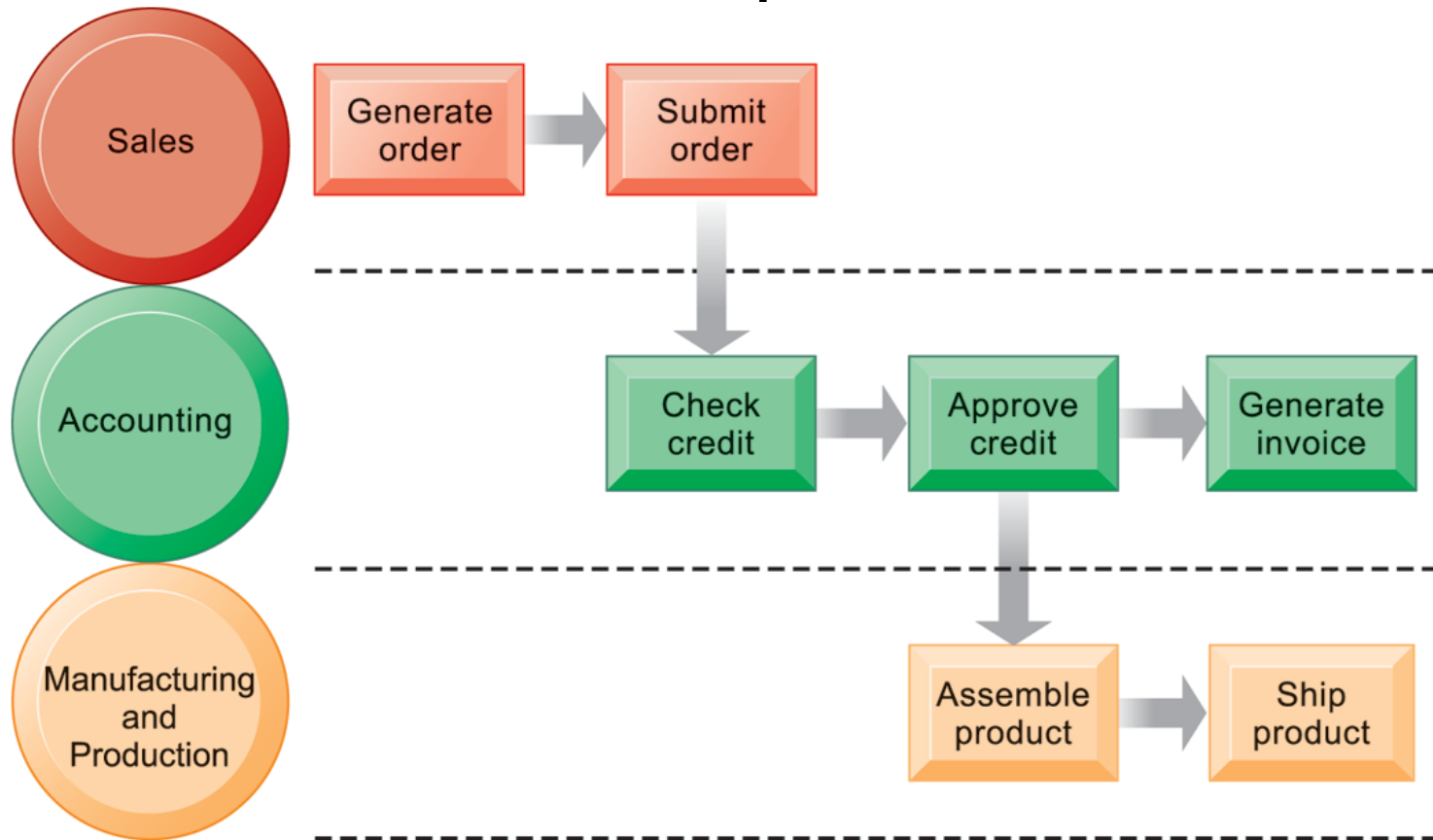
Business process

- The *manner* in which work is organized, coordinated, and focused to produce a valuable product/service
- The *collection of activities* required to produce a product/service
 - Supported by flows of material, information, and knowledge among the participants

Source: wikipedia

Example of a business process

□ The order fulfillment process



Information system development process

- Similar to software development process
 - Information systems are expressed through application software
- Specially focus to the end-user factor
 - End-user: a person who performs tasks of a business process
 - Business: make value for business, improve performance

Information system development tools

- ☐ Tools support following stages
 - ☐ Requirement collection and analysis
 - ☐ Business process analysis
 - ☐ Data analysis
 - ☐ Information system analysis and design
 - ☐ Testing
 - ☐ Deployment
 - ☐ Operations/Maintenance

Components of an IS



Hardware



Software



Telecommunications

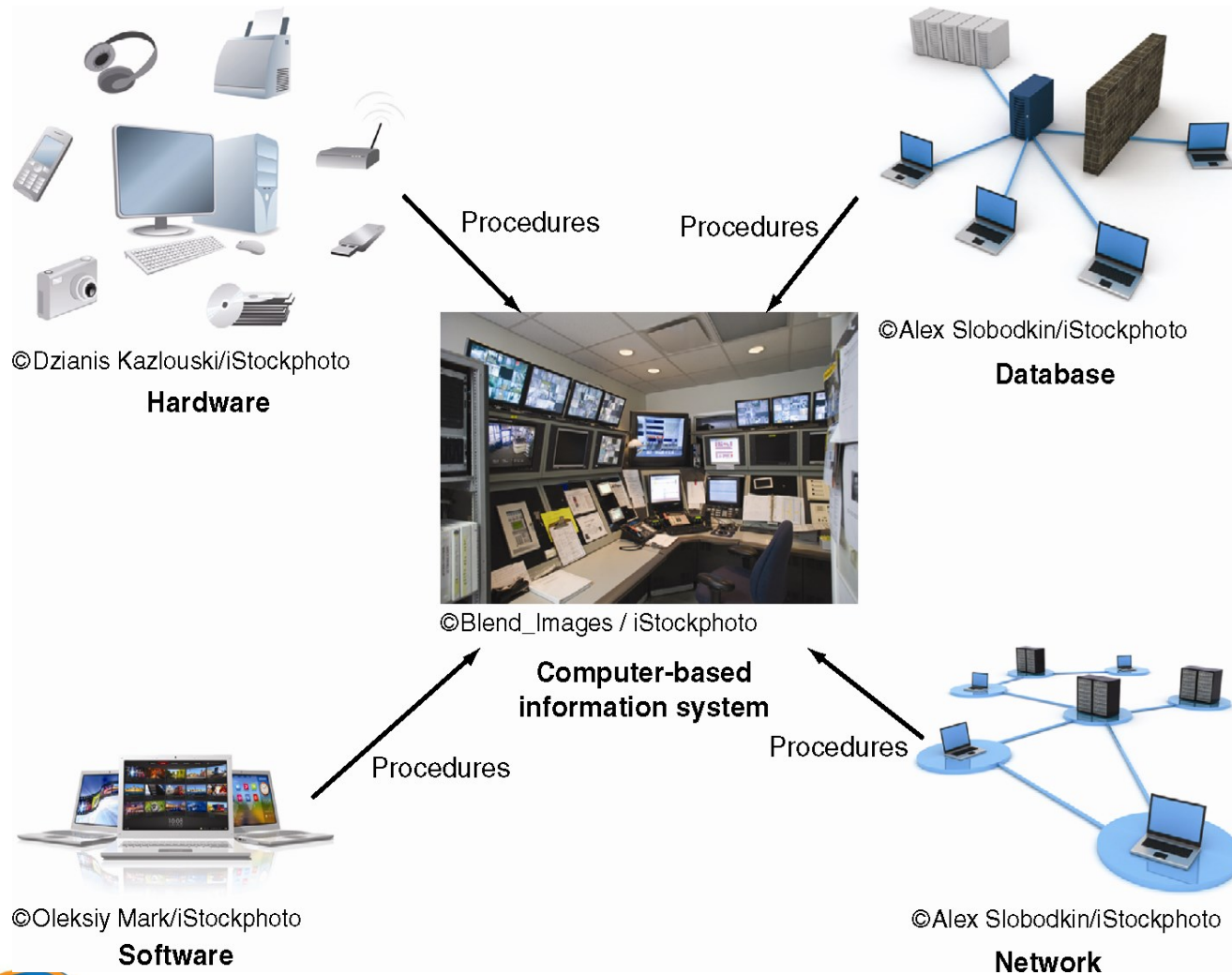


People



Procedures

Components of an IS



Hardware, Software, and Databases

- Hardware: computer equipment used to perform input, processing, storage, and output activities
- Software: computer programs that govern the operation of the computer
- Database: organized collection of facts and information
 - Typically consists of two or more related data files

Telecommunications, Networks, and the Internet

- Telecommunications: electronic transmission of signals for communications
- Networks connect computers and equipment enabling electronic communication
- Internet: world's largest computer network
 - Thousands of interconnected networks

Cloud-Computing and the Web

- Cloud-computing environment:
 - Provides software and data storage via the Internet in order to:
 - Run services on another organization's computer hardware
 - Easily access software and data
- World Wide Web (WWW or Web)
 - Network of links on the Internet to documents containing text, graphics, video, and sound

- Intranet: internal network that allows people within an organization to exchange information and work on projects
- Extranet: network based on Web technologies
 - Allows selected outsiders, e.g., business partners and customers, to access authorized resources of a company's intranet

Business Information System (BIS)

- A group of interrelated components that work collectively to carry out input, processing, output, storage and control actions in order to convert data into information products that can be used to support forecasting, control, coordination, decision making and operational activities in an organization.

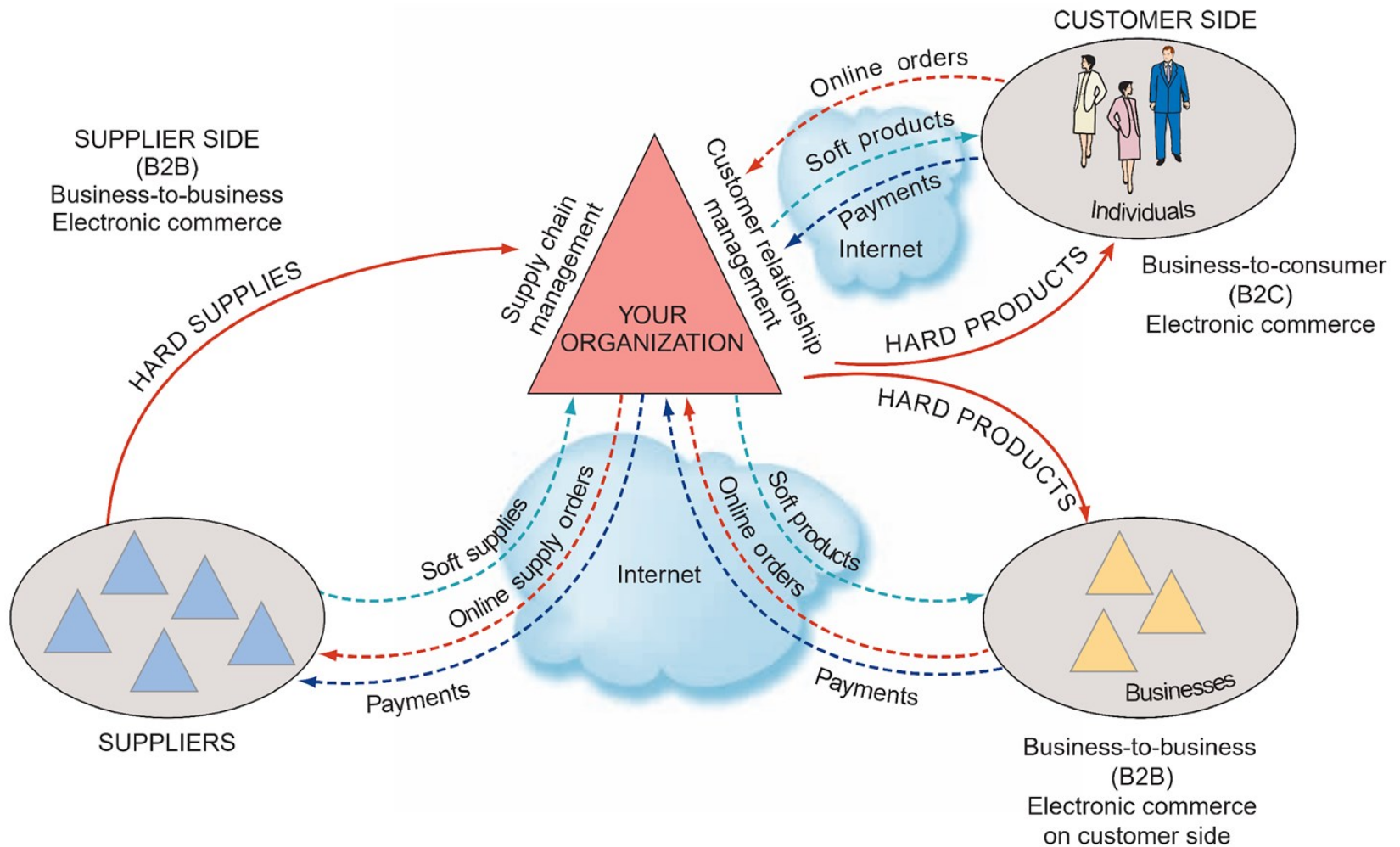
- Most common types of information systems:
 - Systems for electronic and mobile commerce, transaction processing, management information, and decision support

- Special-purpose systems are utilized by organisations

Why BIS important?

- Operational Excellence
 - Improve the efficiency of operations in order to achieve higher profitability. E.g. having the correct amount of stock in store so consumers can always get what they want
- New product services and business models
 - IS systems play a major role for businesses in creating new products and services. New business models can be created and these can describe how a company produce, create and sell there products.
- Customer and Supplier intimacy
 - The better services a company provides its consumers with more likely they are too come back to them and as result the more they will buy off the supplier therefore creating a good relationship with both parties.

BIS - Example



Why BIS important?

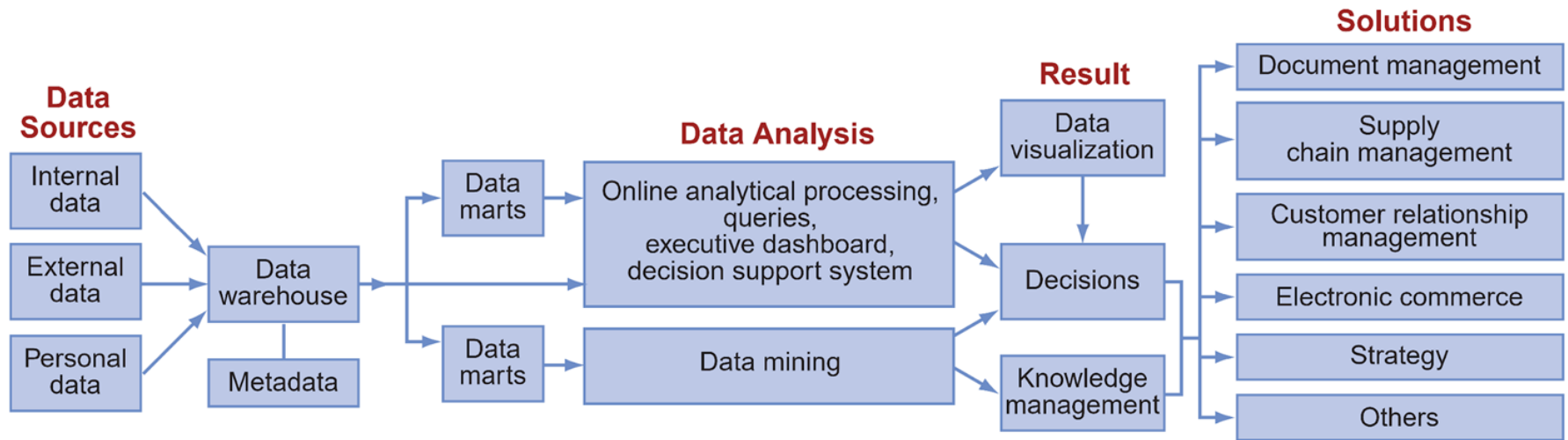
- Improved decision making
 - Managers can use real time data to make better decisions and not have to waste time looking for information.
- Competitive advantage
 - IS systems will generally create a competitive advantage over their rivals
- Day to Day survival
 - Business invest in these systems to make their jobs as easy as possibly.
 - Example: is Citibank introduced the first ATM machine to make it easier for customers to access their money and to cut down queues in their banks.

Major branches of BIS

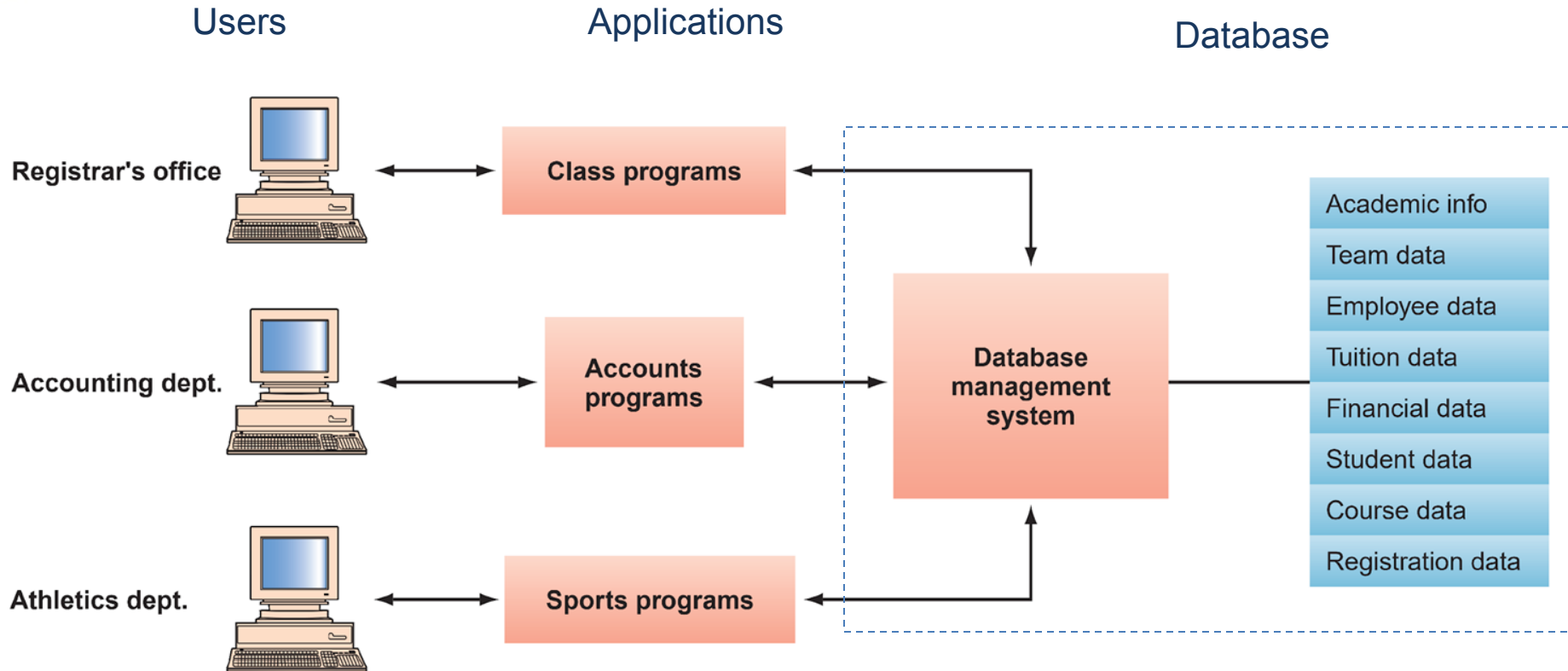


Database Fundamentals

Data Life Cycle



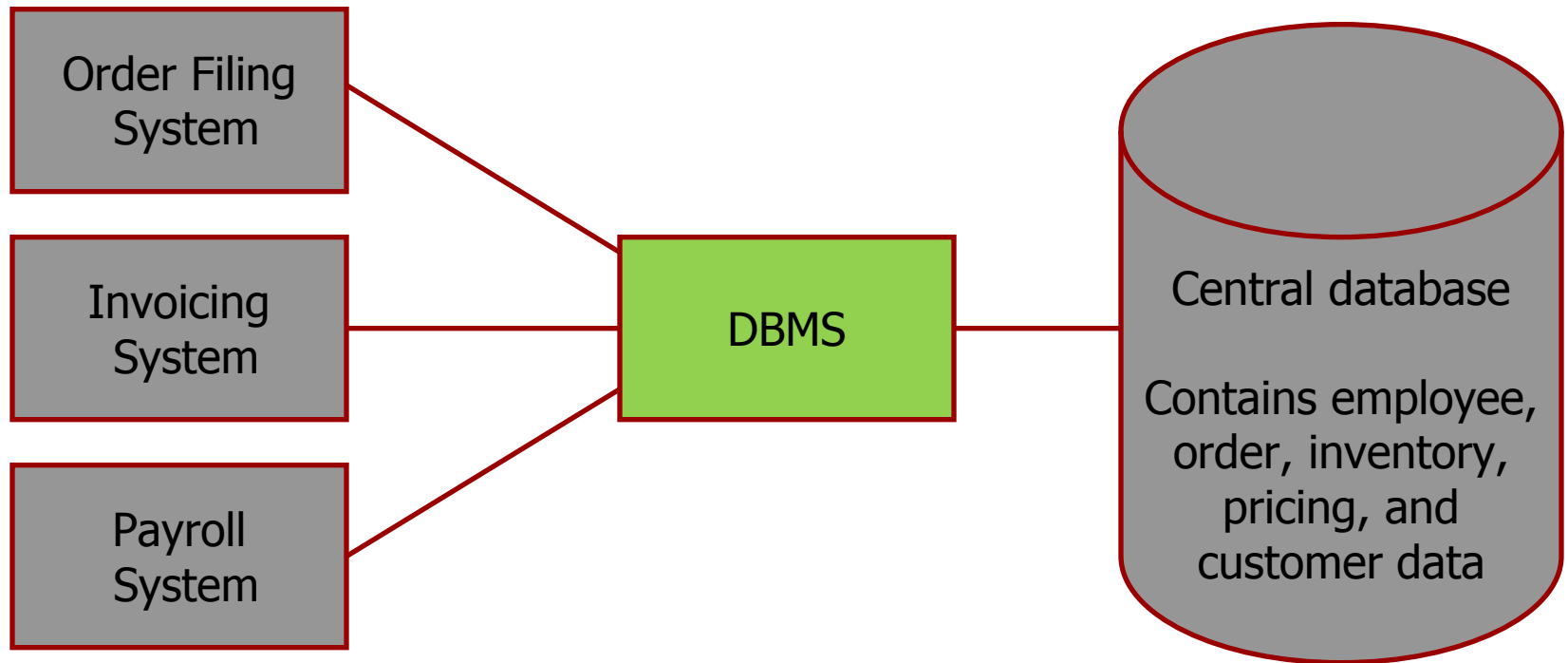
Database Approach



A database management system (DBMS) provides access to all data in the database.

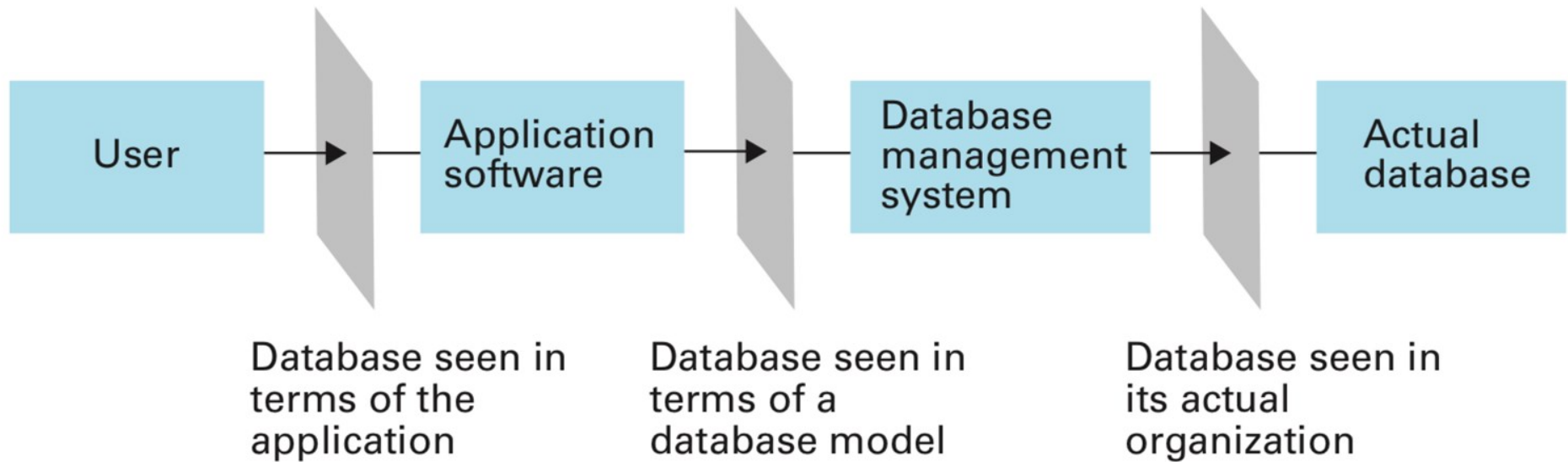
Database Management System

A software system that is used to create, maintain, and provide controlled access to user databases



DBMS manages data resources like an operating system manages hardware resources

Conceptual layers of a database implementation



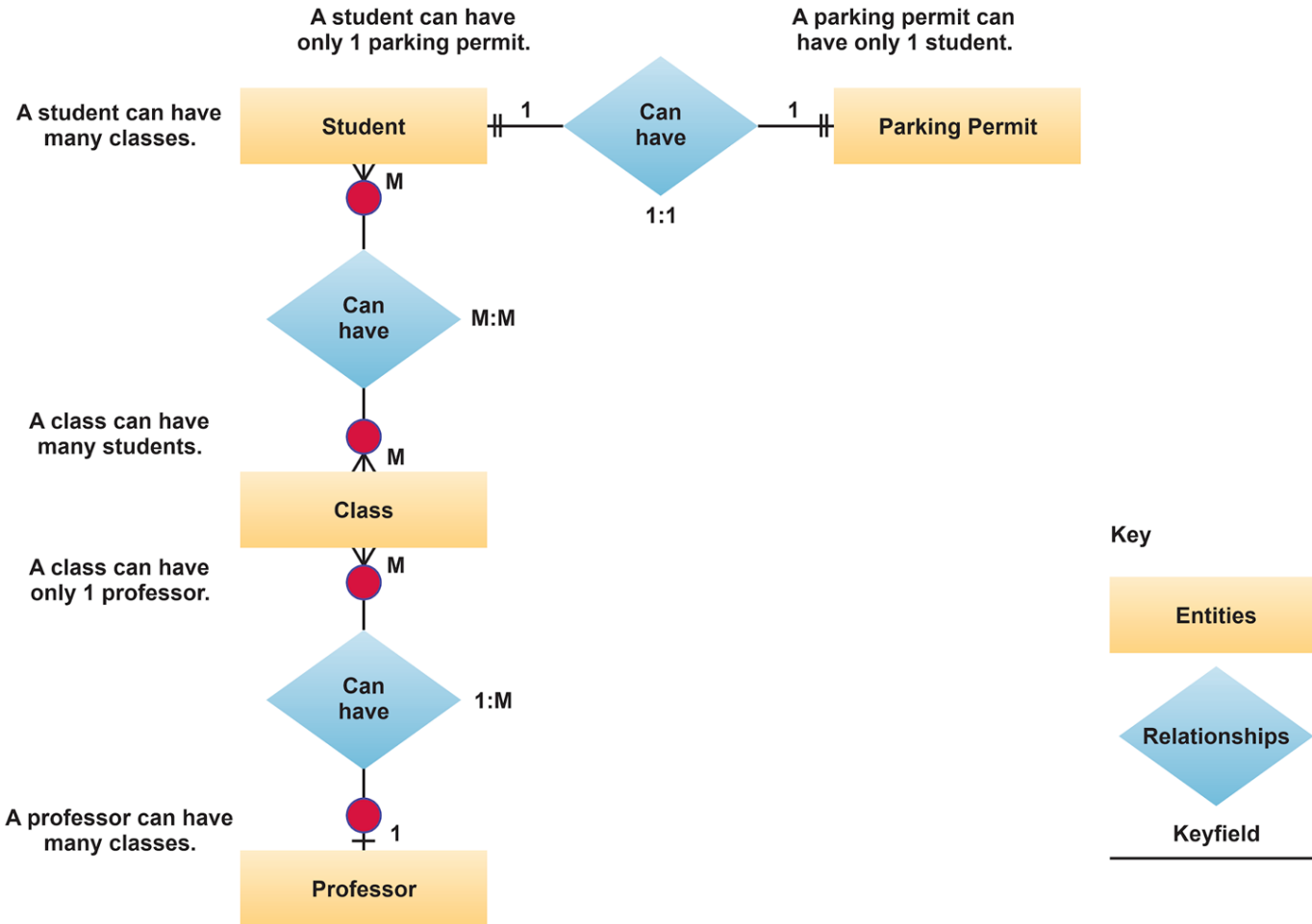
From Brookshear, 2015

Database models

- A conceptual view of a database
 - Entity Relationship Model (ERM)
 - Relational Model
 - Object-oriented Model
 - Hierarchical Model
 - NoSQL
 - ...

- ERM, Relational model is the most popular model

Entity Relationship Model



(a) ER diagram

Relational database model

CUSTOMER

<u>CustomerID</u>	CustomerName	CustomerAddress	CustomerCity*	CustomerState*	CustomerPostalCode
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Primary Key

ORDER

<u>OrderID</u>	OrderDate	<u>CustomerID</u>
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Foreign Key

(implements 1:N relationship between customer and order)

ORDER LINE

<u>OrderID</u>	<u>ProductID</u>	OrderedQuantity
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Combined, these are a *composite primary key* (uniquely identifies the order line)...individually they are *foreign keys* (implement M:N relationship between order and product)

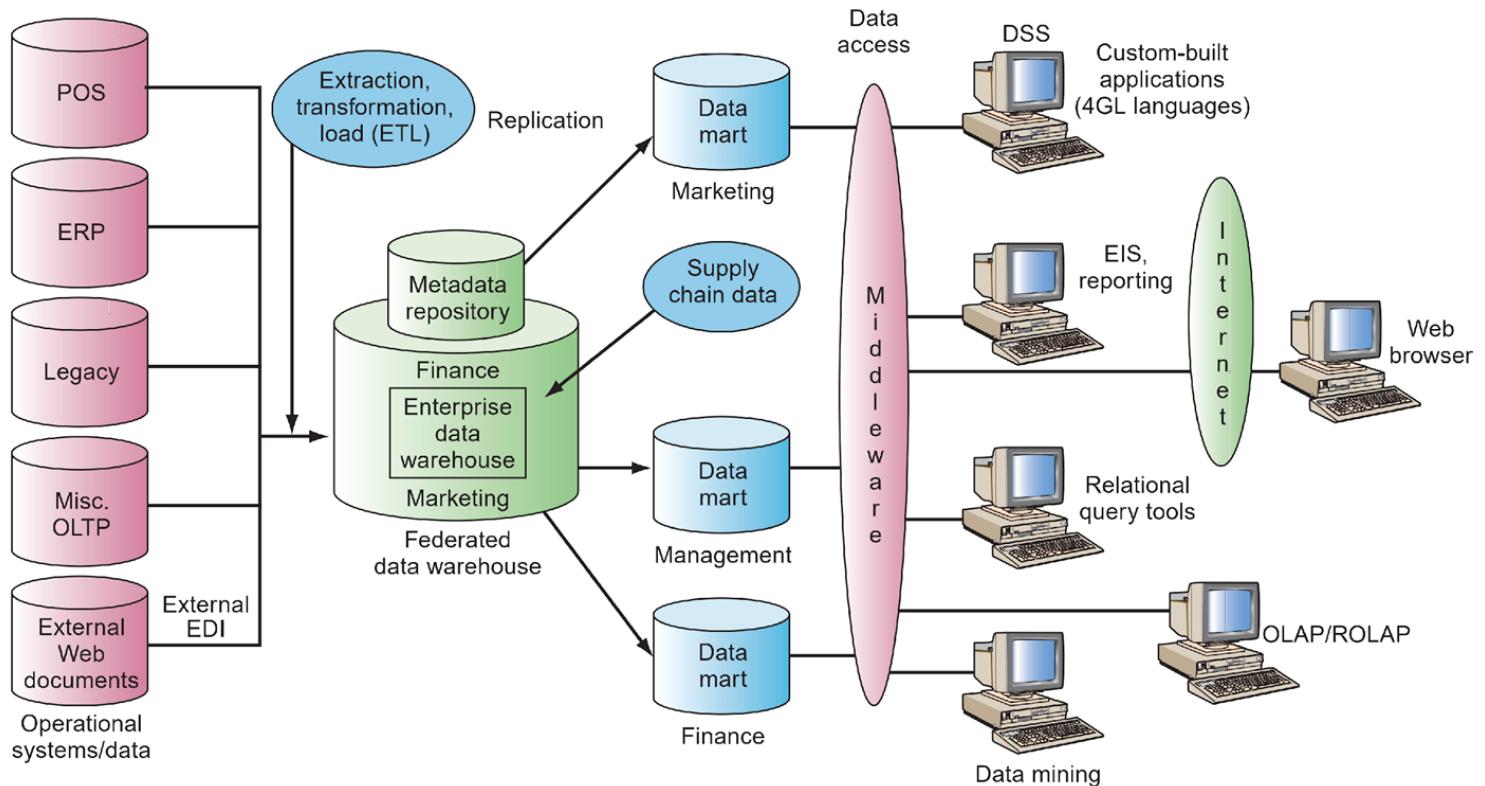
PRODUCT

<u>ProductID</u>	ProductDescription	ProductFinish	ProductStandardPrice	ProductLineID
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Datawarehouse

Data warehouse (DW or DWH), also known as an enterprise data warehouse (EDW), is a system used for reporting and data analysis, and is considered a core component of business intelligence. DWs are central repositories of integrated data from one or more disparate sources. (Wikipedia)

Business intelligence



Data warehouse framework and views

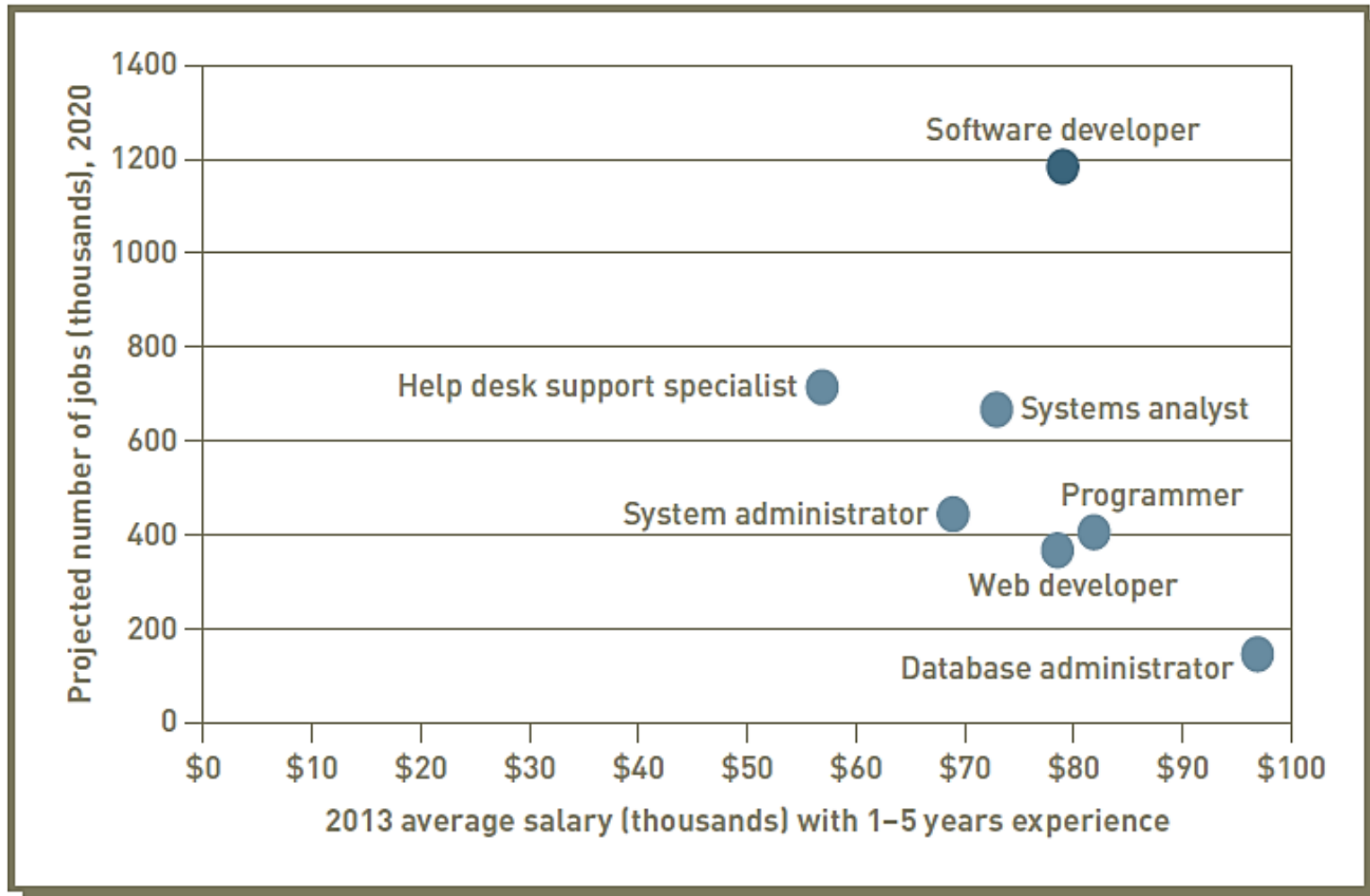
Careers in Information Systems

- Successful IS workers must:
 - Enjoy working in a fast-paced, dynamic environment
 - Meet deadlines and solving unexpected challenges
 - Possess good communication skills
 - Have solid analytical and decision-making skills
 - Develop effective team and leadership skills
 - Be adept at implementing organization change
 - Be prepared to engage in life-long learning

Important Skills for IS Workers

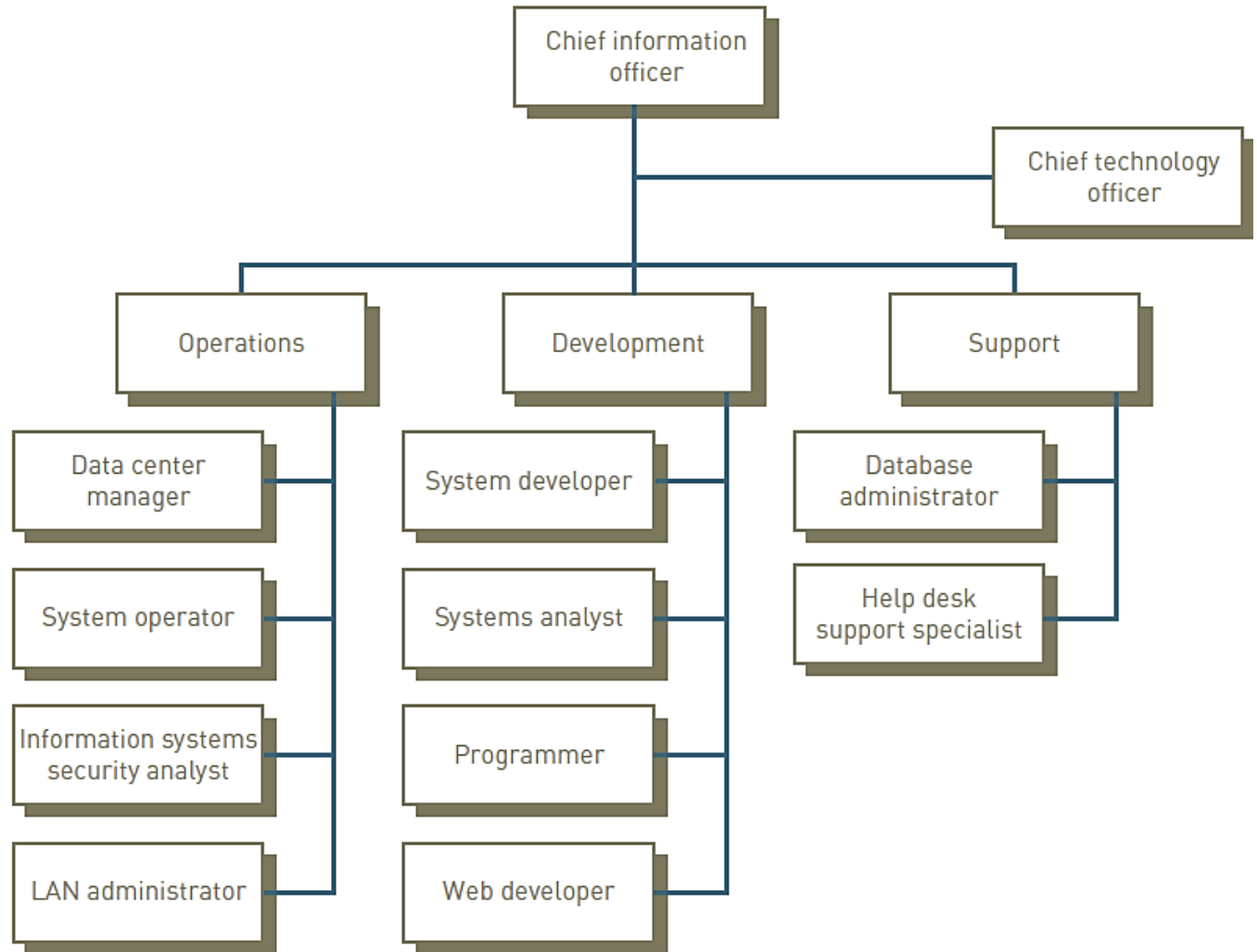
- ☐ Mobile applications for smartphones, tablet computers, and other mobile devices
- ☐ Programing and application development
- ☐ Help desk and technical support
- ☐ Project management
- ☐ Networking
- ☐ Business intelligence
- ☐ Security
- ☐ Web 2.0, 3.0
- ☐ Data center

Occupational Outlook for Selected Information Systems Positions



- In addition to technical skills, IS professionals need:
 - Communication: skills in written and verbal communication
 - Problem solving
 - An understanding of organizations and business and the way they operate
 - Team work: ability to work with people and in groups
 - Etc.

Three Primary Functions of the Information Systems Organization



Typical IS Titles and Functions

- Chief Information Officer (CIO) employs the IS department's equipment and personnel to help the organization attain its goals
- Senior IS Managers
 - IS Director
 - Vice President of Information Systems
 - Manager of Information Systems
 - Chief Technology Officer (CTO)
 - Information Management Manager

Typical IS Titles and Functions: Operations Roles

- Data center managers:
 - Responsible for the maintenance and operation of the organization's computing facilities
- Operations group includes:
 - Operations Manager
 - System Operators
 - Information System Security Analysts

Typical IS Titles and Functions: Development Roles

- The development group implements the new information systems required to support the organization's existing and future business
- The development group includes:
 - IS Project Manager
 - Systems Analyst (SA), Business Analyst (BA)
 - Database designer, Web designer
 - Programmers
 - Web developers

Typical IS Titles and Functions: Support

- Provides customer service for the employees, customers, and business partners

- The support group includes:
 - Database Administrators (DBAs)
 - Help Desk Support Specialists

IS Careers in Technology Organizations

- ☐ Consulting opportunities
- ☐ Computer training
- ☐ Computer and computer-equipment sales
- ☐ Computer repair and maintenance
- ☐ Many other related careers

